Contoso Database Analysis :
Unlocking Business Intelligence with SQL

INDEX

INTRODUCTION

The ***Contoso database*** is a comprehensive sample dataset commonly used to demonstrate data analysis, reporting, and Business Intelligence (BI) techniques. It simulates a real-world retail business scenario, containing structured data on **customers**, **products**, **sales transactions**, **currency exchanges**, stores and **calendar dates**.

This presentation explores how meaningful insights can be derived from this data using **SQL queries**. By examining patterns across sales, customer behavior, product performance, and currency exchange trends, we demonstrate how Business Intelligence helps in making data-driven decisions that can improve business operations and strategy.

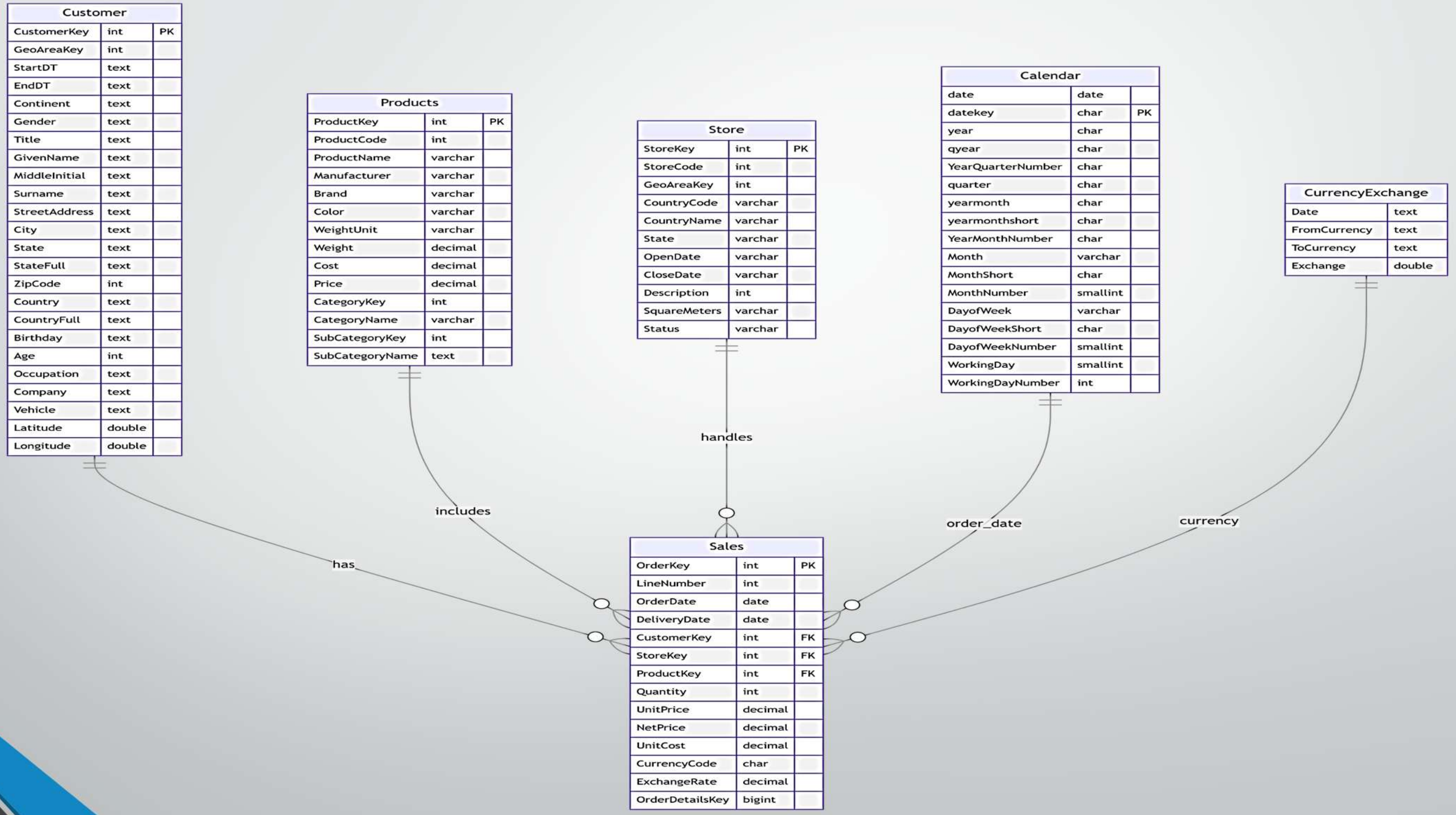
TOPIC DETAILS

- This presentation focuses on analyzing the **Contoso sample database**, which represents a fictional retail company's business operations. The database contains multiple interconnected tables including:
- **Calendar** – Provides date-related information used for time-based analysis.
- **CurrencyExchange** – Stores exchange rates between currencies over time.
- **Customer** – Contains customer details useful for segmentation and behavior analysis.
- **Products** – Lists product information such as names, categories, and prices.
- **Sales** – Records transactions linking customers, products, dates, and revenue.
- **Store** – Has all the records of products, customer data.

Through a series of SQL queries, this project demonstrates how to:

- Filter and retrieve meaningful records (e.g., weekend sales, Monday transactions).
- Join tables to combine data from multiple business areas.
- Group and summarize data to gain insights.
- Apply Business Intelligence techniques to support decision-making.
- The goal is to show how raw business data can be transformed into valuable insights using SQL — the foundation of Business Intelligence.

EER DIAGRAM



Customer
CustomerKey int PK
GeoAreaKey int
StartDT text
EndDT text
Continent text
Gender text
Title text
GivenName text
MiddleInitial text
Surname text
StreetAddress text
City text
State text
StateFull text
ZipCode int
Country text
CountryFull text
Birthday text
Age int
Occupation text
Company text
Vehicle text
Latitude double
Longitude double
Products
ProductKey int PK
ProductCode int
ProductName varchar
Manufacturer varchar
Brand varchar
Color varchar
WeightUnit varchar
Weight decimal
Cost decimal
Price decimal
CategoryKey int
CategoryName varchar
SubCategoryKey int
SubCategoryName text
Store
StoreKey int PK
StoreCode int
GeoAreaKey int
CountryCode varchar
CountryName varchar
State varchar
OpenDate varchar
CloseDate varchar
Description int
SquareMeters varchar
Status varchar
Calendar
date date
datekey char PK
year char
qyear char
YearQuarterNumber char
quarter char
yearmonth char
yearmonthshort char
YearMonthNumber char
Month varchar
MonthShort char
MonthNumber smallint
DayofWeek varchar
DayofWeekShort char
DayofWeekNumber smallint
WorkingDay smallint
WorkingDayNumber int
CurrencyExchange
Date text
FromCurrency text
ToCurrency text
Exchange double
handles
includes
has
order_date
currency
Sales
OrderKey int PK
LineNumber int
OrderDate date
DeliveryDate date
CustomerKey int FK
StoreKey int FK
ProductKey int FK
Quantity int
UnitPrice decimal
NetPrice decimal
UnitCost decimal
CurrencyCode char
ExchangeRate decimal
OrderDetailsKey bigint

1. Get all records in the ‘calendar’ where ‘WorkingDay’ = 1.

```
SELECT * FROM calendar WHERE WorkingDay = 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows: | Result Grid

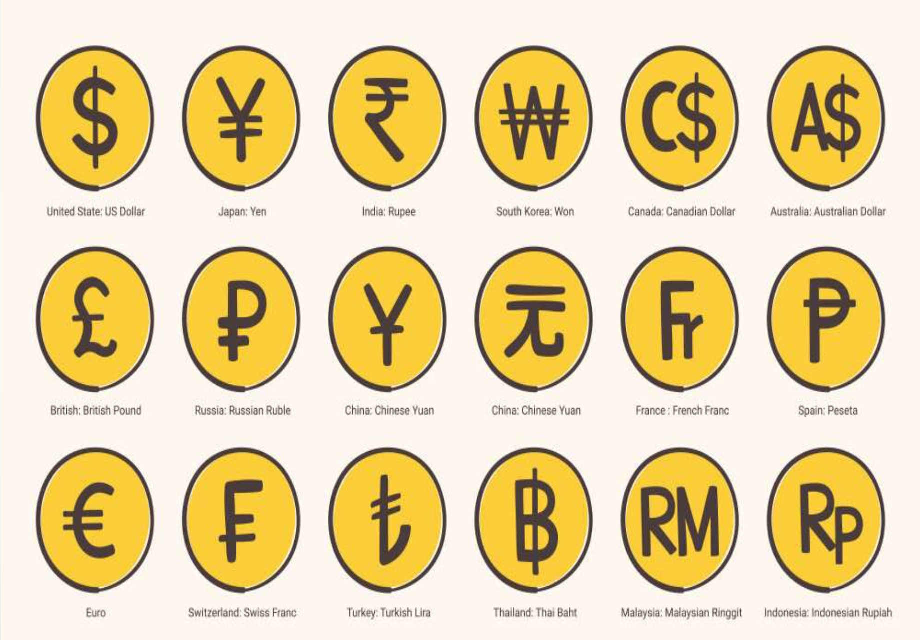
date	datekey	year	qyear	YearQuarterNumber	quarter	yearmonth	yearmonthshort	YearMonthNumber	Month	MonthShort	MonthNumber	DayofWeek	DayofWeekShort	DayofWeekNumber	WorkingDay	
2015-01-02	20150102	2015	Q1-2015	8061	Q1	Jan-15	Jan-15	24181	January	Jan	1	Friday	Fri	6	1	1
2015-01-05	20150105	2015	Q1-2015	8061	Q1	Jan-15	Jan-15	24181	January	Jan	1	Monday	Mon	2	1	2
2015-01-06	20150106	2015	Q1-2015	8061	Q1	Jan-15	Jan-15	24181	January	Jan	1	Tuesday	Tue	3	1	3
2015-01-07	20150107	2015	Q1-2015	8061	Q1	Jan-15	Jan-15	24181	January	Jan	1	Wednesday	Wed	4	1	4

1 06:33:53 SELECT * FROM calendar WHERE WorkingDay = 1 2509 row(s) returned 0.203 sec / 0.015 sec

2. How to retrieve exchange rates where `FromCurrency` is not equal to `ToCurrency`.

```sql
SELECT * FROM currencyexchange
WHERE FromCurrency != ToCurrency ;
```

	Date	FromCurrency	ToCurrency	Exchange
▶	2015-01-01	AUD	CAD	0.94834
	2015-01-01	AUD	EUR	0.67435
	2015-01-01	AUD	GBP	0.52525
	2015-01-01	AUD	USD	0.81873
	2015-01-01	CAD	AUD	1.05447



3 06:50:06 select * from currencyexchange -- WHERE FromCurrency != ToCurrency WHERE FromCurrency <> ToCurrency 73060 row(s) returned 0.250 sec / 0.437 sec

3.How can we get the number of working days per month ?

```
SELECT    Year,    Month,    MonthNumber,    COUNT(*) AS WorkingDays
FROM calendar
WHERE WorkingDay = 'Yes’
GROUP BY Year, Month, MonthNumber
ORDER BY Year, MonthNumber;
```

Year	Month	MonthNumber	WorkingDays
2015	January	1	11
2015	February	2	9
2015	March	3	9
2015	April	4	8
2015	May	5	11

5 07:05:43 SELECT Year, Month, MonthNumber, COUNT(*) AS WorkingDays FROM calendar WHERE WorkingD... 120 row(s) returned 0.188 sec / 0.000 sec

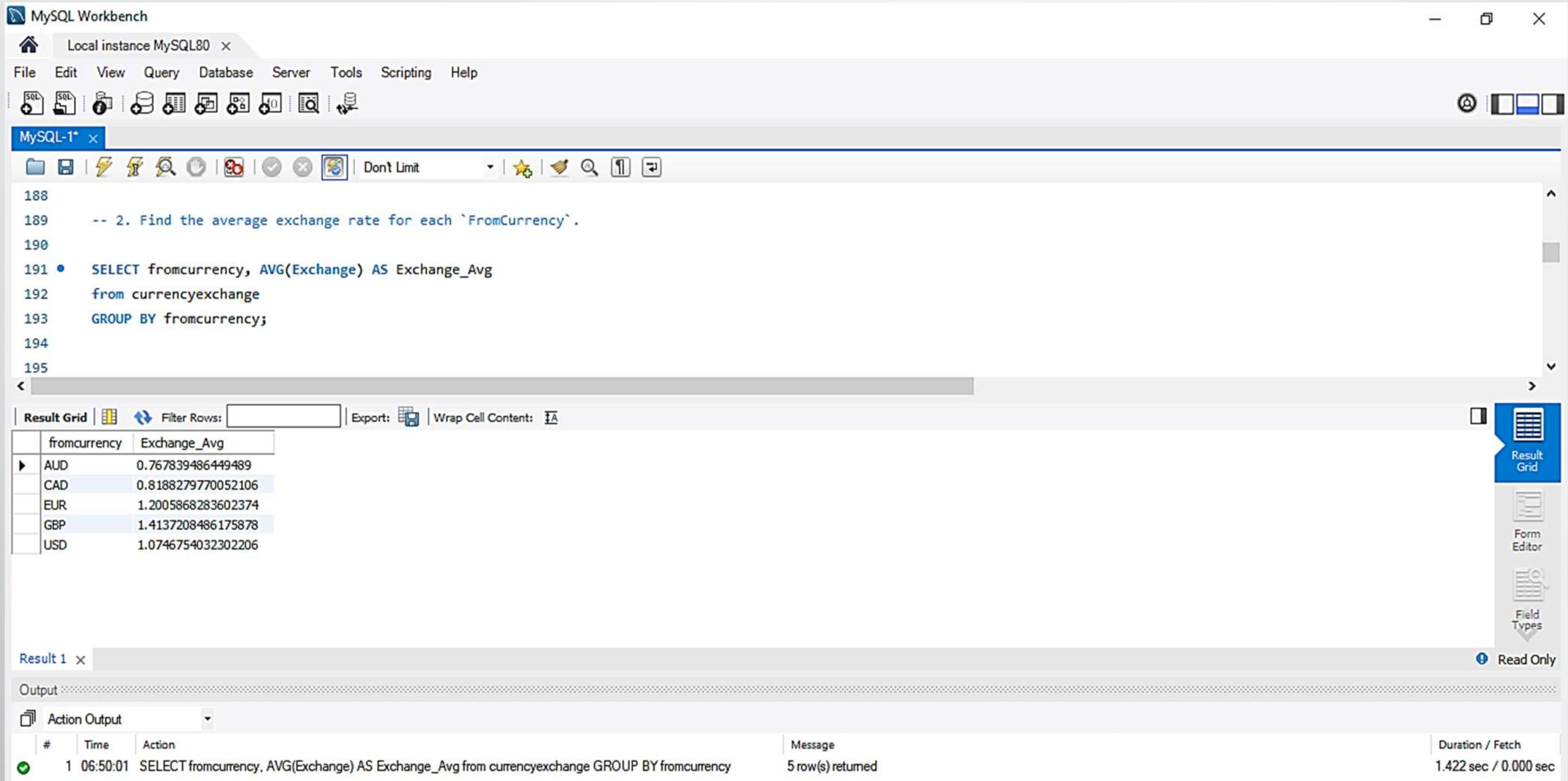
4. How to find the sum of 'WorkingDayNumber' per 'MonthShort'.

```
SELECT year, monthshort, SUM(WorkingDayNumber) AS workingdayNo_sum
FROM calendar
GROUP BY Year, MonthShort;
```

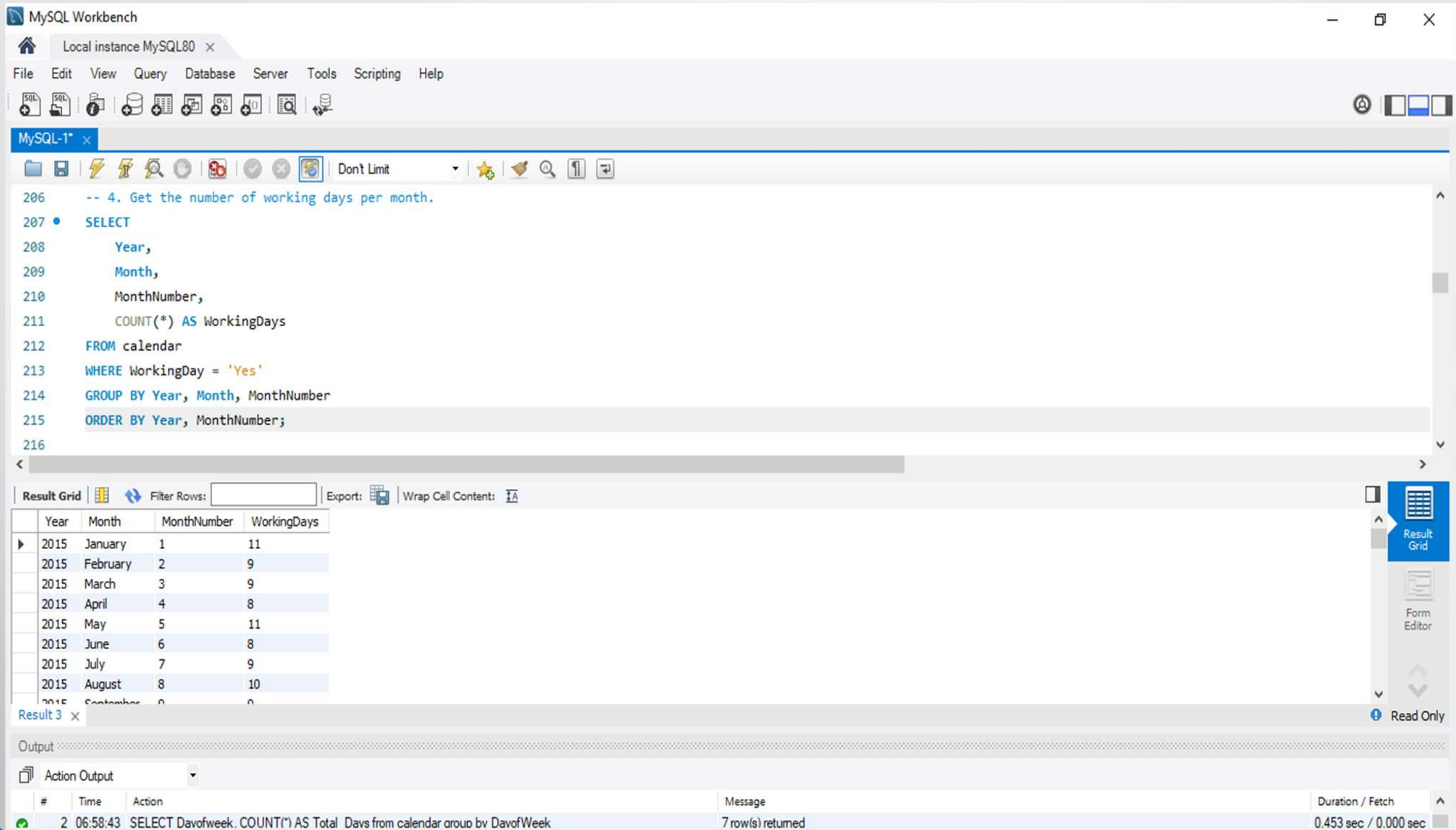
year	monthshort	workingdayNo_sum
2015	Jan	307
2015	Feb	837
2015	Mar	1562
2015	Apr	2167
2015	May	2907

23 07:25:28 select year, monthshort, SUM(WorkingDayNumber) AS workingdayNo_sum from calendar GROUP BY Year, Mont... 120 row(s) returned 0.172 sec / 0.000 sec

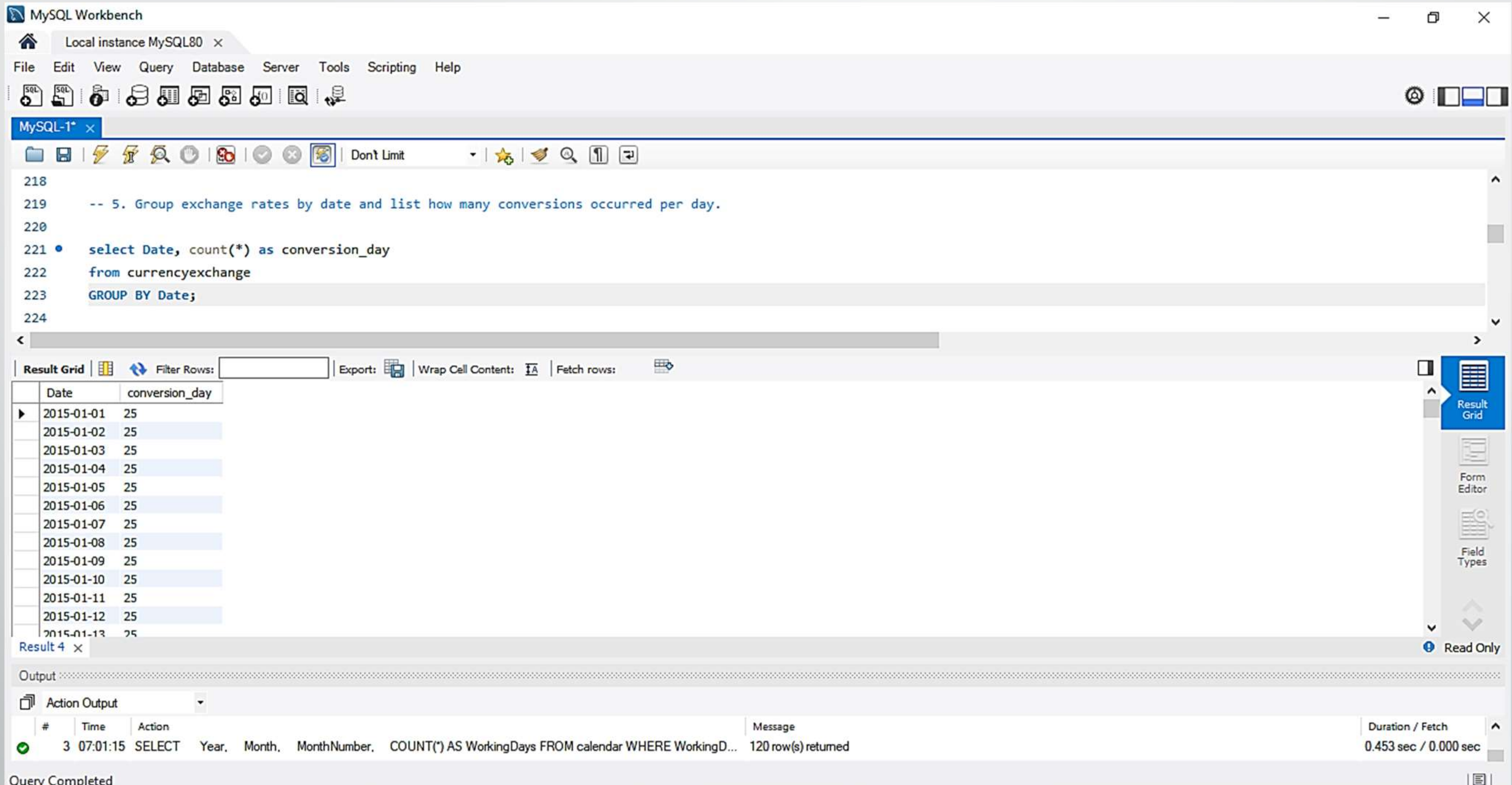
5. How to Find the average exchange rate for each 'FromCurrency"



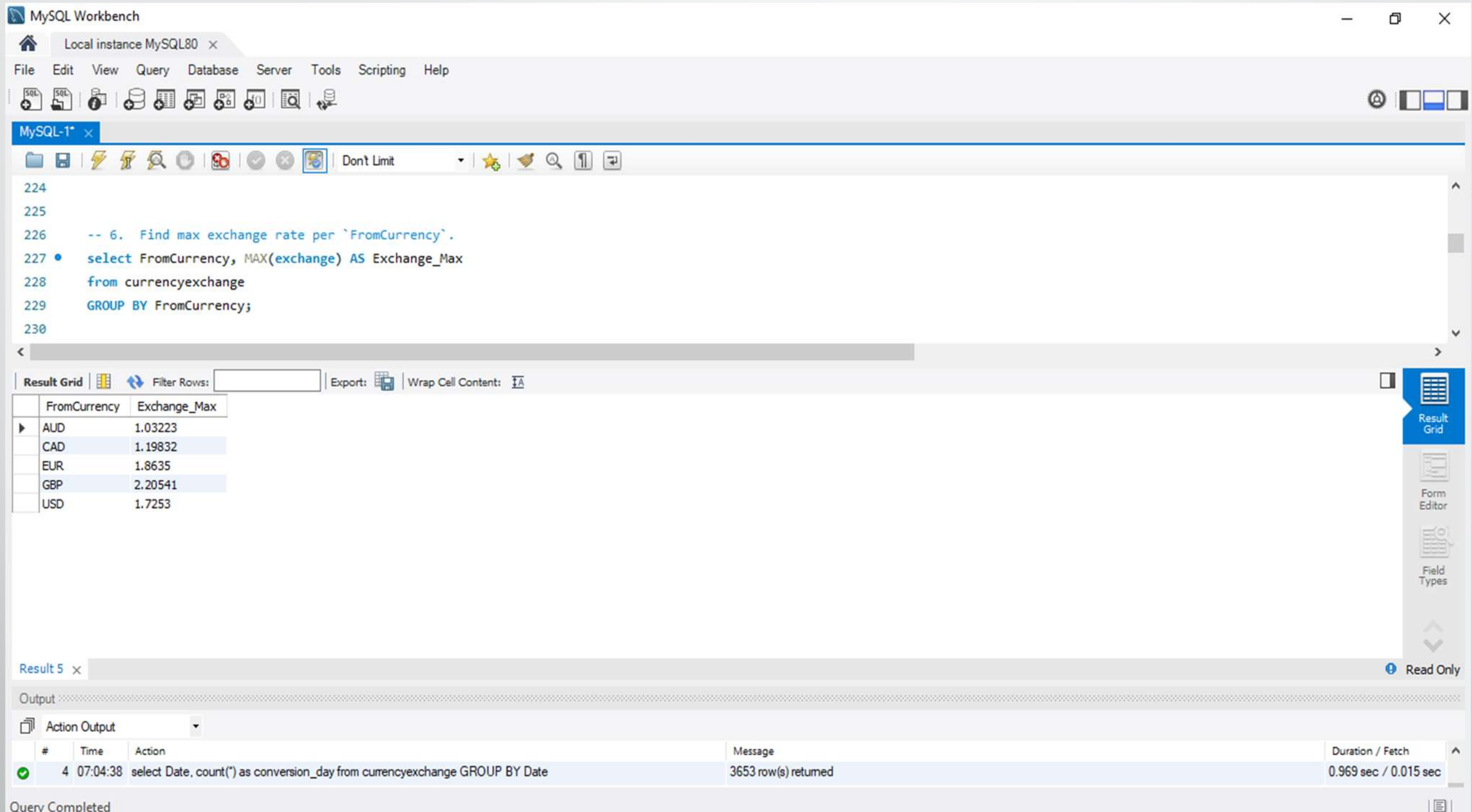
6. How to get number of working days



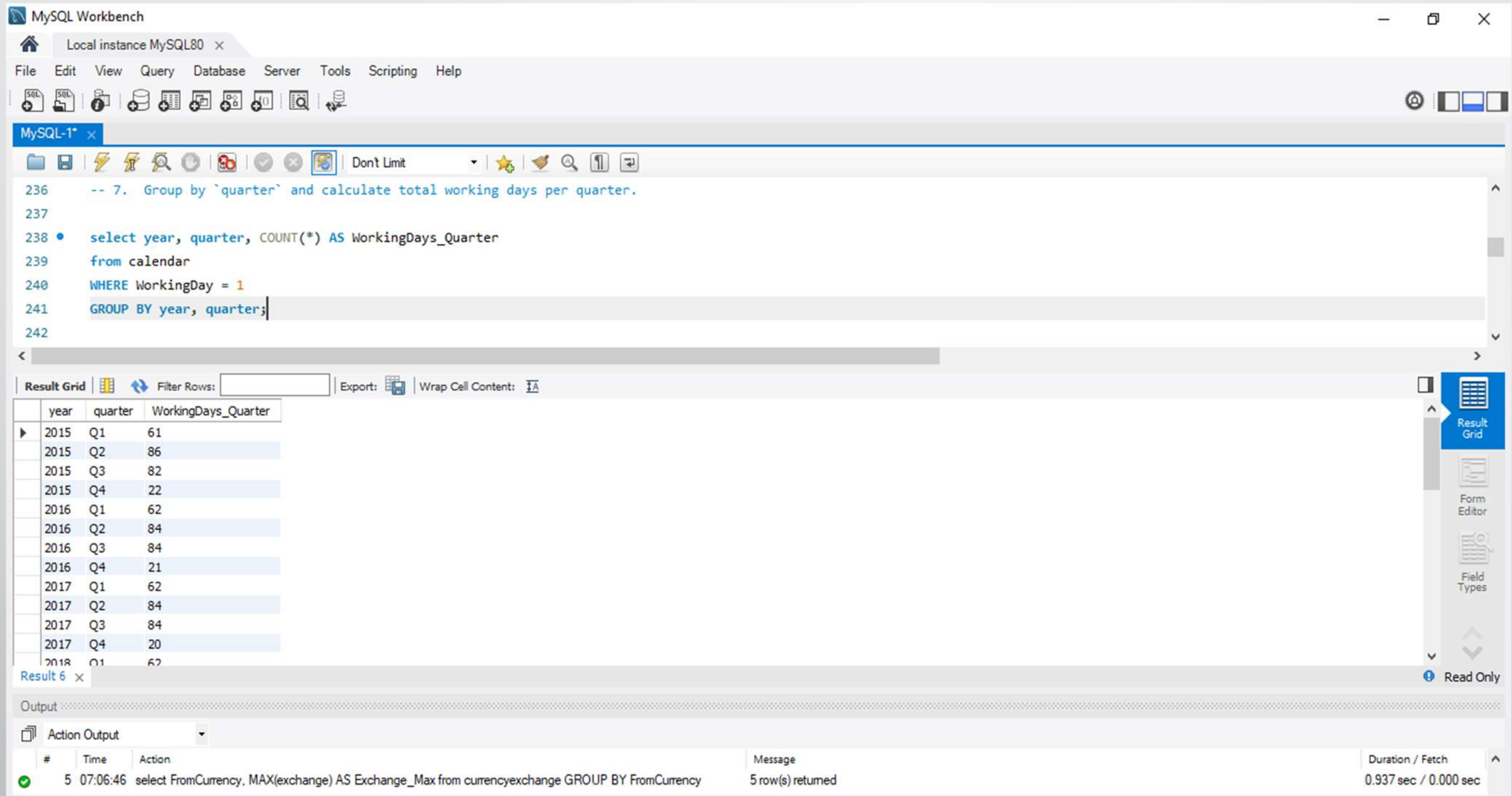
7. How to get exchange rates by date and list how many conversions



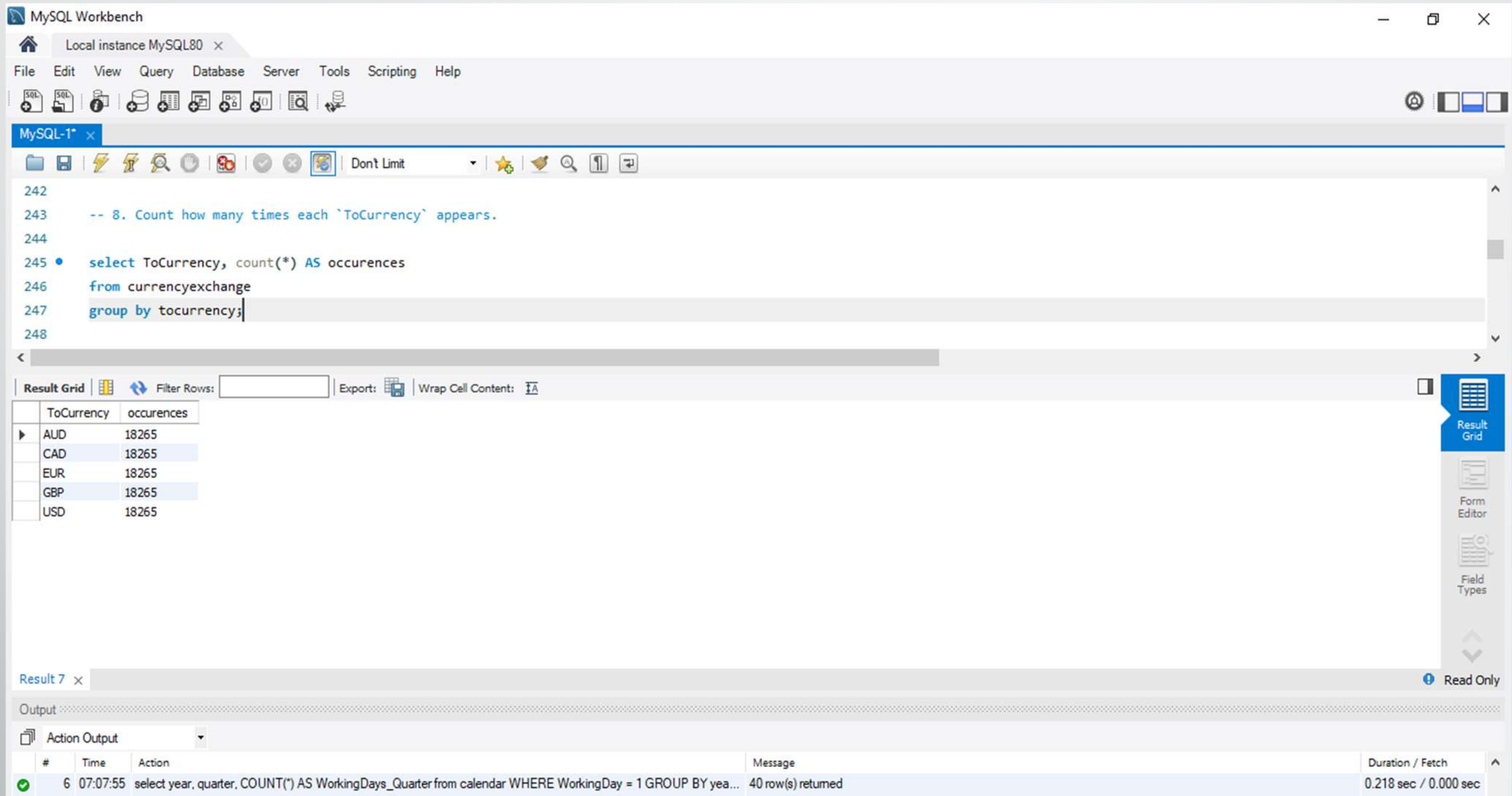
8. Find max exchange rate per `FromCurrency`.



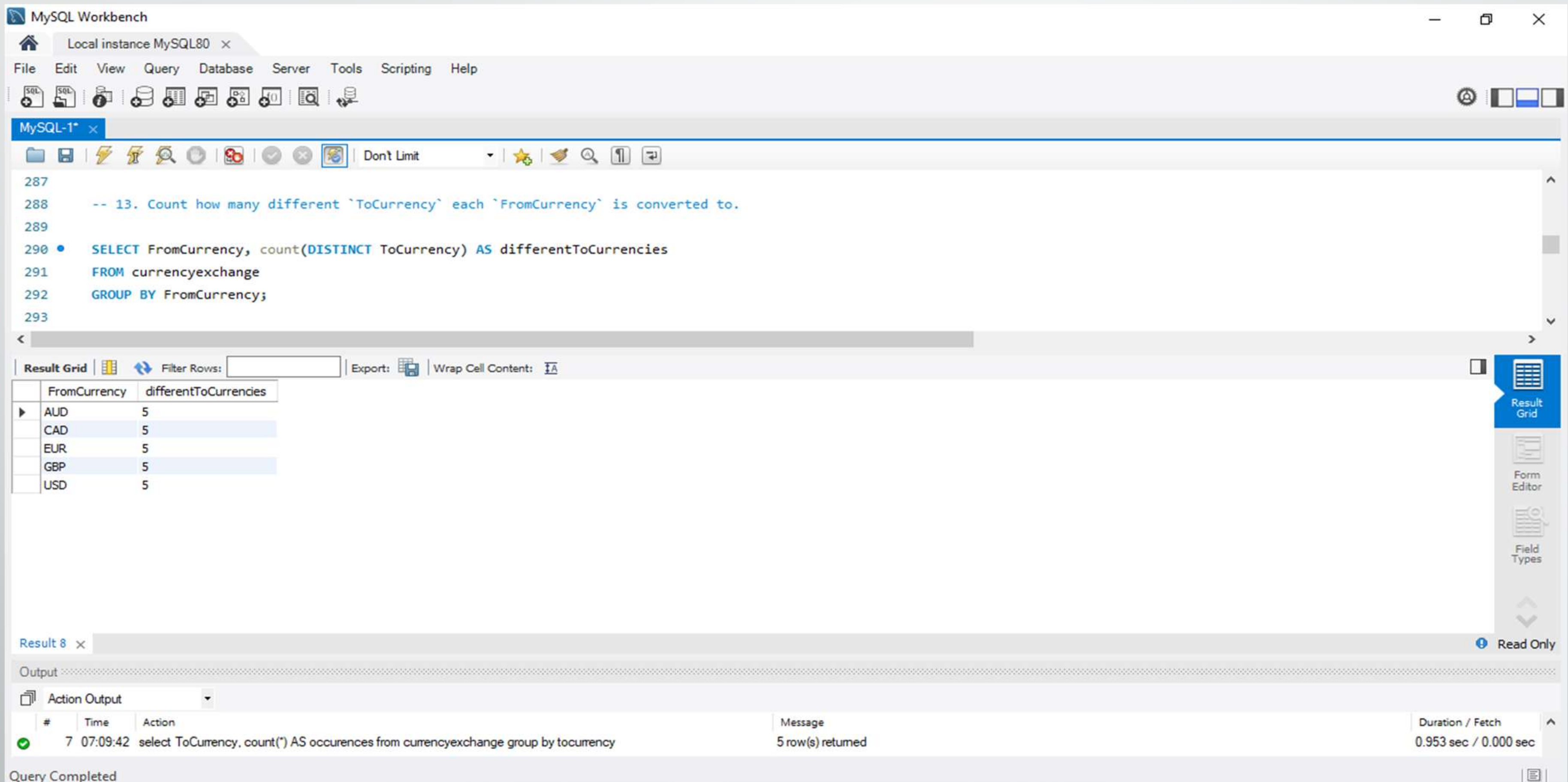
9. Group by 'quarter' and calculate total working days per quarter.



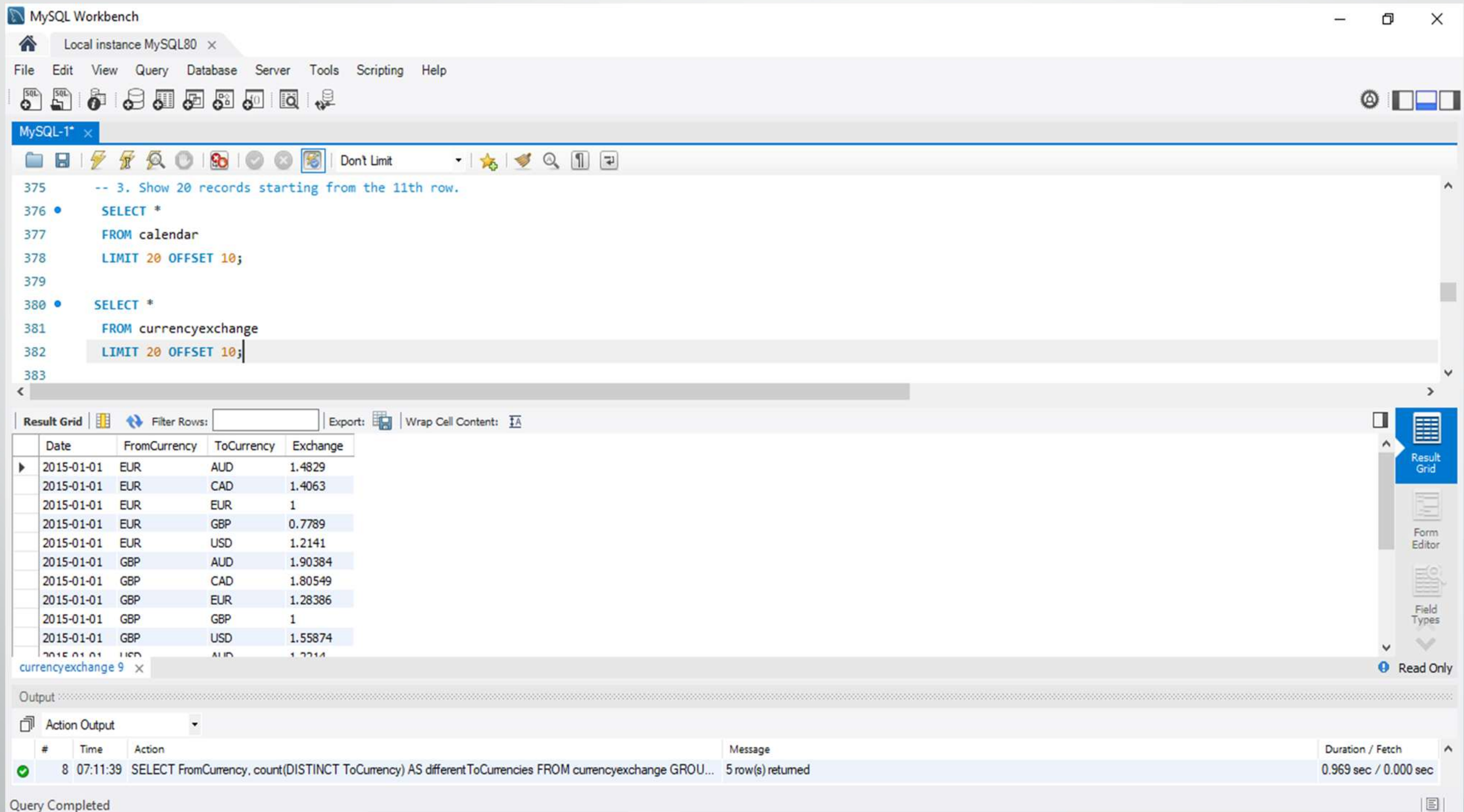
10. Count how many times each `ToCurrency` appears.



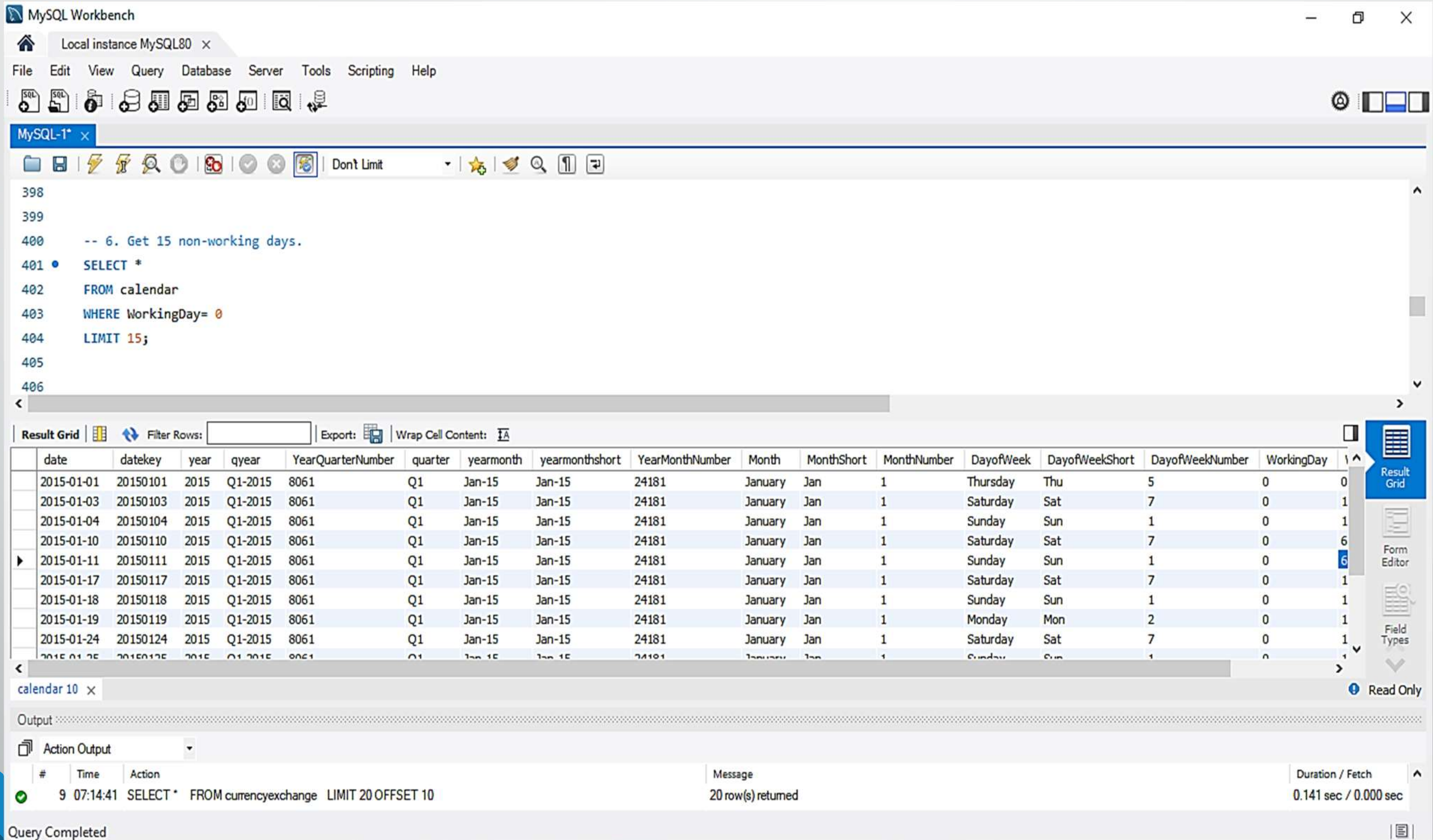
11. Count how many different `ToCurrency` each `FromCurrency` is converted to.



12. Show 20 records starting from the 11th row.



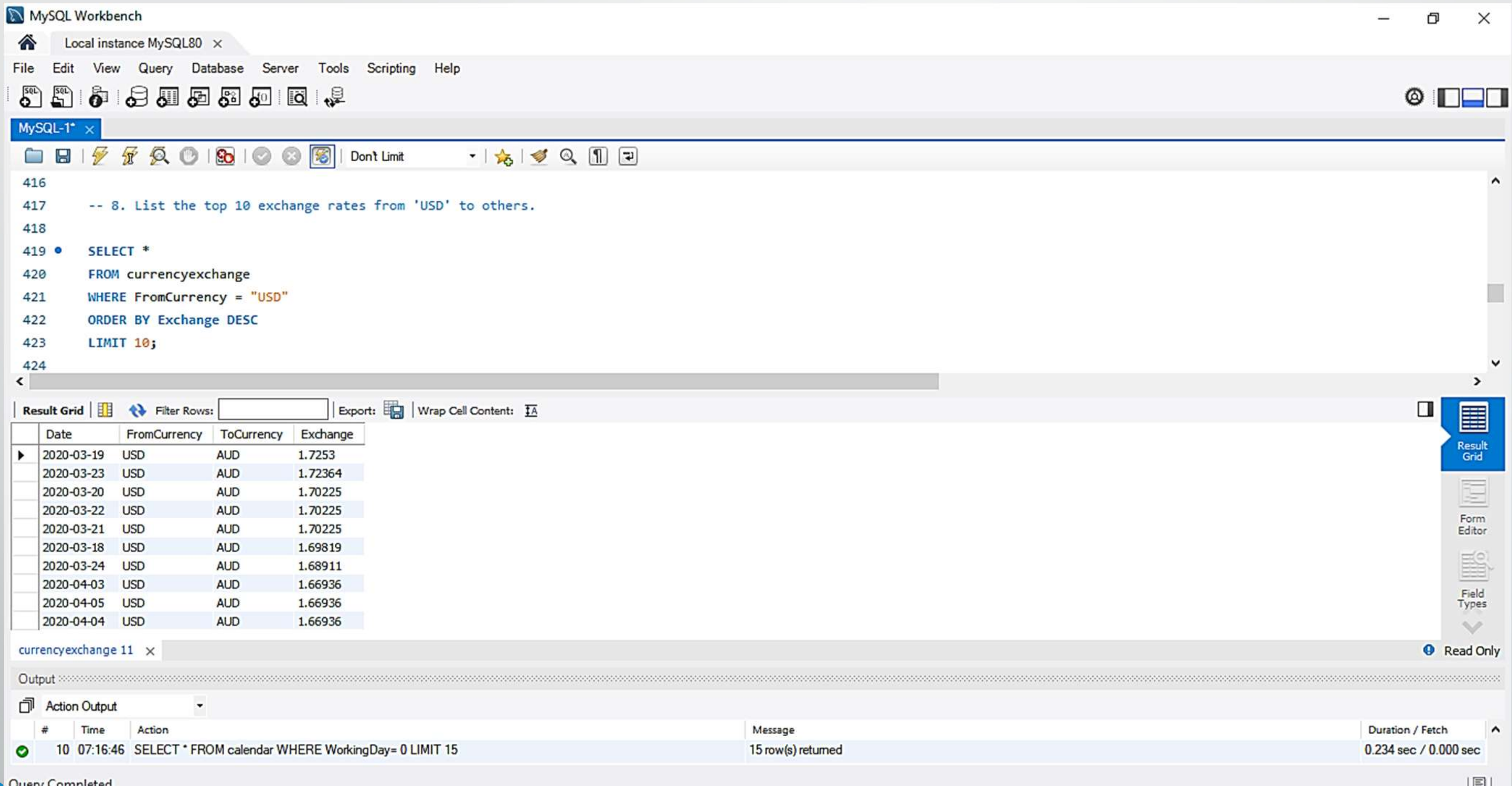
13. How to get 15 non-working days.



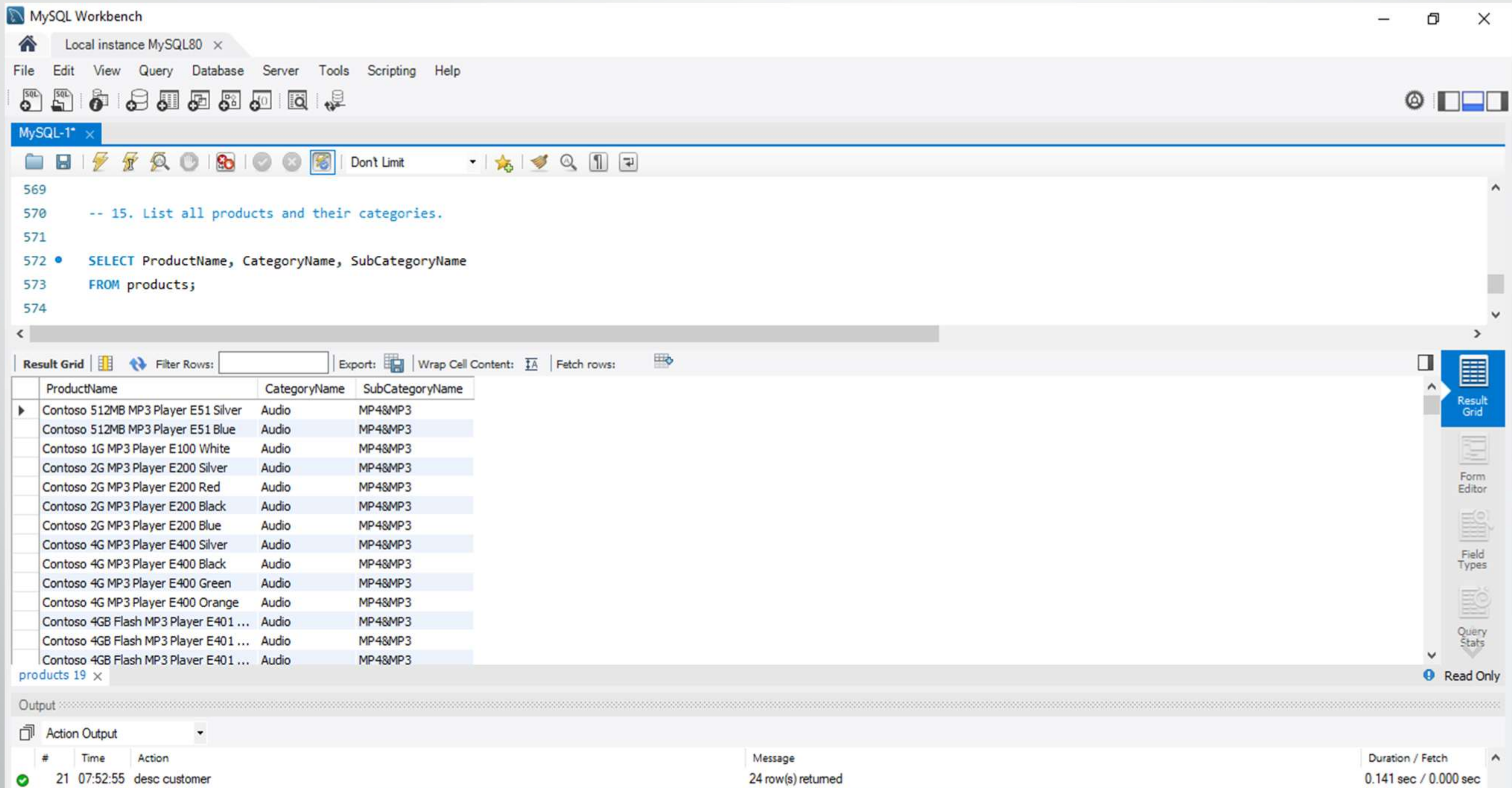

```sql
-- 6. Get 15 non-working days.
SELECT *
FROM calendar
WHERE WorkingDay= 0
LIMIT 15;
```

date	datekey	year	qyear	YearQuarterNumber	quarter	yearmonth	yearmonthshort	YearMonthNumber	Month	MonthShort	MonthNumber	DayofWeek	DayofWeekShort	DayofWeekNumber	WorkingDay	V
2015-01-01	20150101	2015	Q1-2015	8061	Q1	Jan-15	Jan-15	24181	January	Jan	1	Thursday	Thu	5	0	0
2015-01-03	20150103	2015	Q1-2015	8061	Q1	Jan-15	Jan-15	24181	January	Jan	1	Saturday	Sat	7	0	1
2015-01-04	20150104	2015	Q1-2015	8061	Q1	Jan-15	Jan-15	24181	January	Jan	1	Sunday	Sun	1	0	1
2015-01-10	20150110	2015	Q1-2015	8061	Q1	Jan-15	Jan-15	24181	January	Jan	1	Saturday	Sat	7	0	6
2015-01-11	20150111	2015	Q1-2015	8061	Q1	Jan-15	Jan-15	24181	January	Jan	1	Sunday	Sun	1	0	6
2015-01-17	20150117	2015	Q1-2015	8061	Q1	Jan-15	Jan-15	24181	January	Jan	1	Saturday	Sat	7	0	1
2015-01-18	20150118	2015	Q1-2015	8061	Q1	Jan-15	Jan-15	24181	January	Jan	1	Sunday	Sun	1	0	1
2015-01-19	20150119	2015	Q1-2015	8061	Q1	Jan-15	Jan-15	24181	January	Jan	1	Monday	Mon	2	0	1
2015-01-24	20150124	2015	Q1-2015	8061	Q1	Jan-15	Jan-15	24181	January	Jan	1	Saturday	Sat	7	0	1

14. List the top 10 exchange rates from 'USD' to others.



15. List all products and their categories.



16. Get customers who have never placed an order.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

MySQL-1*

Don't Limit

```sql
-- Get customers who have never placed an order.
SELECT *
FROM customer c
WHERE c.CustomerKey NOT IN (
    SELECT DISTINCT CustomerKey
    FROM sales);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

CustomerKey	GeoAreaKey	StartDT	EndDT	Continent	Gender	Title	GivenName	MiddleInitial	Surname	StreetAddress	City	State	StateFull	ZipCode	Country	CountryFull	Birthc
210	2	1980-09-28	2030-05-08	Australia	female	Mrs.	Natalie	L	Hilder	63 Davenport Street	PADDYS FLAT	NSW	New South Wales	2632	AU	Australia	1950-
322	2	1993-09-27	2031-06-30	Australia	female	Ms.	Alexandra	D	Stephen	61 Hart Street	GREEN CREEK	NSW	New South Wales	2338	AU	Australia	1974-
609	7	1987-09-16	2007-10-13	Australia	female	Mrs.	Lucy	K	Parrott	22 Thomas Lane	BELLFIELD	VIC	Victoria	3081	AU	Australia	1947-
661	2	1985-02-17	2007-11-19	Australia	male	Mr.	Archie	J	Hugh	60 Gaggin Street	KARUAH	NSW	New South Wales	2324	AU	Australia	1994-
827	2	1983-06-07	2042-12-16	Australia	female	Ms.	Erin	B	Kellermann	48 Elgin Street	OAKHAMPTON	NSW	New South Wales	2320	AU	Australia	1980-
855	7	1987-08-03	2018-08-02	Australia	female	Ms.	Kaitlyn	B	Bogle	10 Edmundsons Road	MOLLONGGHIP	VIC	Victoria	3352	AU	Australia	1999-
865	8	1984-06-10	2038-02-06	Australia	male	Mr.	Zane	A	Shellshear	43 Lapko Road	TINGLEDALE	WA	Western Australia	6333	AU	Australia	1978-
928	8	1985-12-08	2009-12-18	Australia	female	Mrs.	Caitlyn	M	Viner	69 Alfred Street	BANDY CREEK	WA	Western Australia	6450	AU	Australia	1937-
977	4	1988-03-16	2038-05-03	Australia	female	Dr.	Mariam	J	Bourke	59 Ross Street	MOUNT NATHAN	QLD	Queensland	4211	AU	Australia	1968-
1259	7	1980-04-03	2013-08-18	Australia	male	Mr.	William	E	Fairfax	54 Gaffney Street	NORLANE	VIC	Victoria	3214	AU	Australia	1963-
1291	5	1986-01-31	2012-08-20	Australia	female	Ms.	Ellie	O	Cayley	11 Gadd Avenue	KALLORA	SA	South Australia	5550	AU	Australia	1990-
1306	6	1991-08-13	2008-04-12	Australia	female	Ms.	Ruby	I	Michell	99 Pelican Road	TOLMANS HILL	TAS	Tasmania	7007	AU	Australia	1989-
1442	8	1989-08-02	2017-05-08	Australia	male	Mr.	Daniel	A	Pelsaert	95 Jacolite Street	SAWYERS VALLEY	WA	Western Australia	6074	AU	Australia	1952-

Result Grid | Form Editor | Field Types | Query Stats

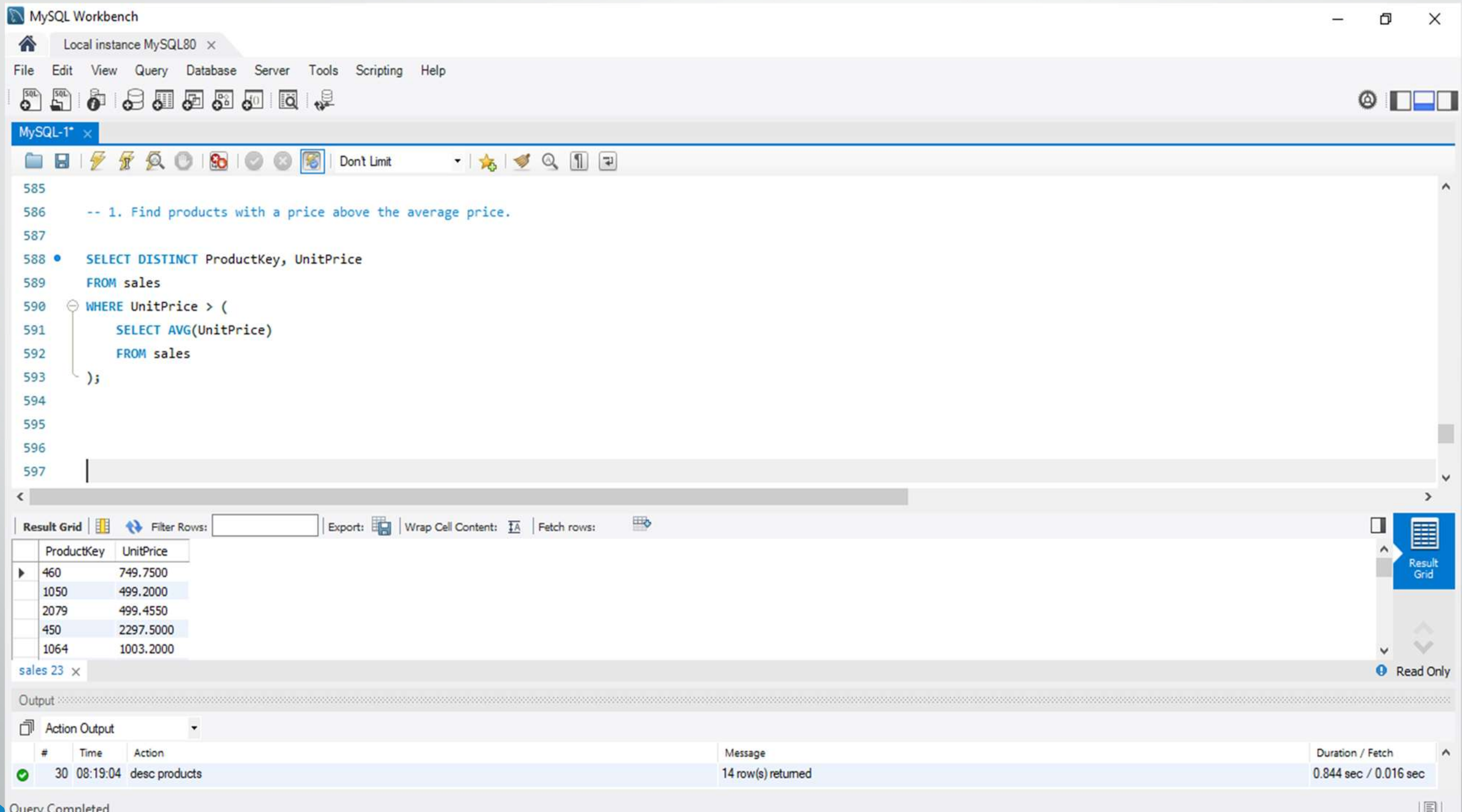
customer 21 × | Read Only

Output

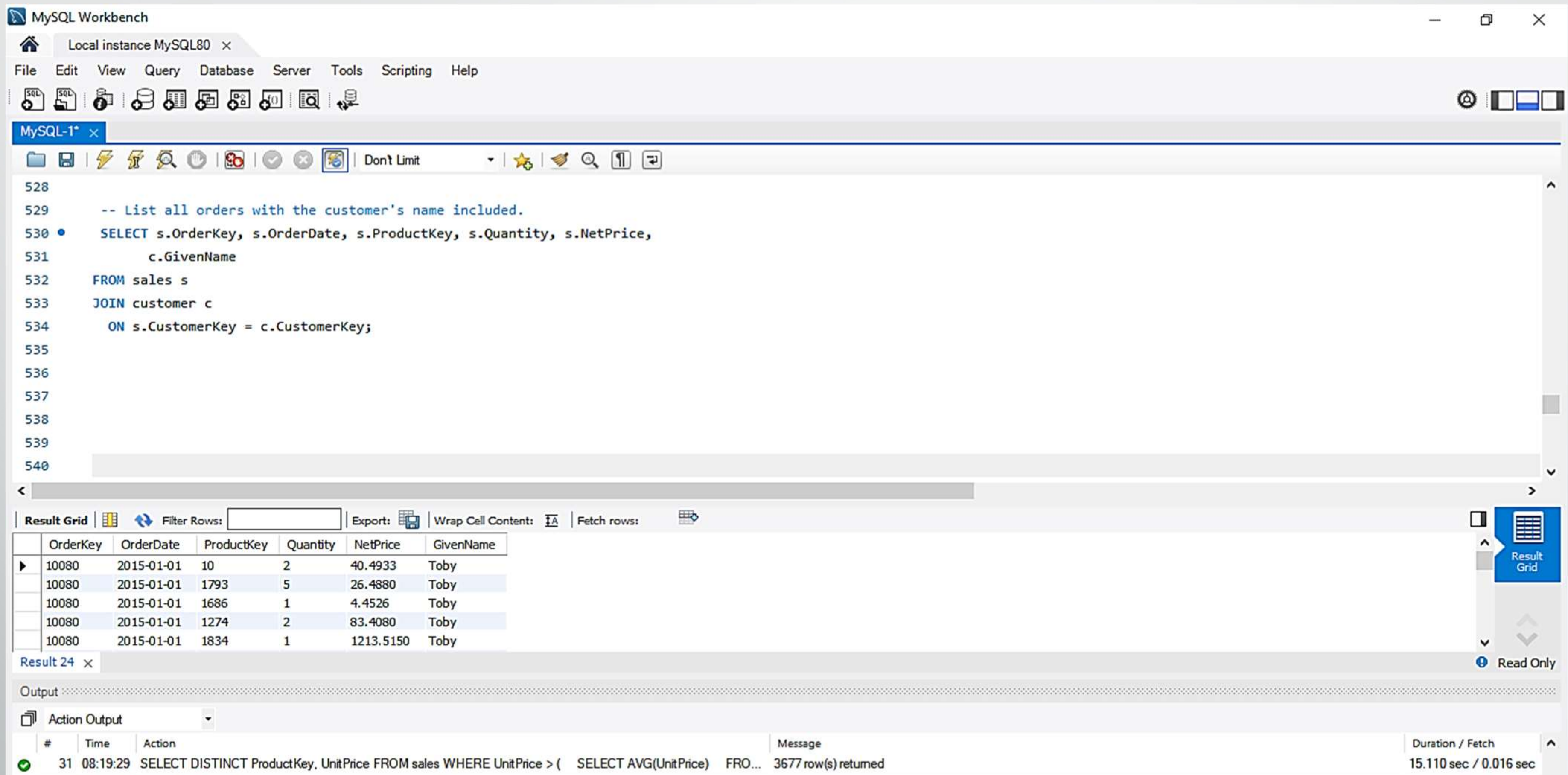
Action Output

#	Time	Action	Message	Duration / Fetch
28	08:15:15	SELECT * FROM customer c WHERE c.CustomerKey NOT IN (SELECT DISTINCT CustomerKey FROM sal...	4503 row(s) returned	23.594 sec / 0.750 sec

17. Find products with a price above the average price.



18. List all orders with the customer's name included.



19. How to retrieve a list of customers and their most recent order date .

```sql
-- Retrieve a list of customers and their most recent order date

SELECT s.orderkey, s.orderdate, s.productkey, s.quantity, s.netprice, c.givenname
FROM sales s
Join customer as c
on s.CustomerKey = c.CustomerKey;
```

orderkey	orderdate	productkey	quantity	netprice	givenname
10080	2015-01-01	10	2	40.4933	Toby
10080	2015-01-01	1793	5	26.4880	Toby
10080	2015-01-01	1686	1	4.4526	Toby
10080	2015-01-01	1274	2	83.4080	Toby
10080	2015-01-01	1834	1	1213.5150	Toby
10080	2015-01-01	1645	1	30.2134	Toby
20014	2015-01-02	139	4	399.9920	Callum
20014	2015-01-02	1271	5	11.1200	Callum
20014	2015-01-02	104	4	75.9000	Callum
20059	2015-01-02	1609	2	137.2747	Dylan
30053	2015-01-03	262	3	148.0560	Dean
30053	2015-01-03	106	3	87.7734	Dean
30090	2015-01-03	825	2	34.7500	Cody
30105	2015-01-03	1426	4	1060.2000	Molly

#	Time	Action	Message	Duration / Fetch
5	06:47:35	SELECT s.orderkey, s.orderdate, s.productkey, s.quantity, s.netprice, c.givenname FROM sales s Join customer as...	420200 row(s) returned	0.125 sec / 5.500 sec

20. How to limit exchange rates to the first 12 EUR-USD conversions.

```sql
-- 19. Limit exchange rates to the first 12 EUR-USD conversions.
SELECT *
FROM currencyexchange
WHERE FromCurrency= "EUR" AND ToCurrency= "USD"
ORDER BY Date ASC
LIMIT 12;
```

Date	FromCurrency	ToCurrency	Exchange
2015-01-01	EUR	USD	1.2141
2015-01-02	EUR	USD	1.2043
2015-01-03	EUR	USD	1.2043
2015-01-04	EUR	USD	1.2043
2015-01-05	EUR	USD	1.1915
2015-01-06	EUR	USD	1.1914
2015-01-07	EUR	USD	1.1831
2015-01-08	EUR	USD	1.1768
2015-01-09	EUR	USD	1.1813
2015-01-10	EUR	USD	1.1813
2015-01-11	EUR	USD	1.1813
2015-01-12	EUR	USD	1.1804

CONCLUSION

- The Contoso Retail Database provides a well-structured sample dataset that represents a real-world retail business. It contains detailed information about sales transactions, products, customers, stores, and time, making it ideal for practicing SQL queries, data analysis, and business intelligence.
- By analyzing this data, we can:
- Understand sales trends over time
- Identify best-selling products and top-performing stores
- Explore customer demographics and buying patterns
- Measure profitability and business performance
- This dataset is especially useful for students and analysts to build practical skills in tools like Power BI, SQL, and Excel, and to simulate real business scenarios for decision-making.

THANK YOU

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